BANK LOAN - DATA ANALYSIS PROJECT

(Bank Loan Report)

Problem Statement: Create a database for bank loan data in MySQL and analyze the various parameters given in the following question. Use **financial_loan.csv** for database creation and refer to other attached files for domain knowledge.

Note: Submission of sql script file is **mandatory** for evaluation.

A. BANK LOAN REPORT | SUMMARY

1. KPI's:

1) Number of Applications
   a) Total Loan Applications

```
select
count(id) as Total_Loan_App
from  financial_loan;
```

Result Output –

Total_Loan_App
38576

b) MTD Loan Applications(Month-To-Date i.e. Current Month)

```
SELECT
   COUNT(*) AS mtd_loan_applications
FROM financial_loan
WHERE YEAR(issue_date) = 2021
  AND MONTH(issue_date) = MONTH(CURDATE());
```

Result Output –

mtd_loan_applications
2627

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c) PMTD Loan Applications(Previous Month)

```
SELECT
MONTHNAME(issue_date) AS Month_Name,
COUNT(*) AS pmtd_loan_app
FROM financial_loan
WHERE MONTH(issue_date) = MONTH(DATE_SUB(CURDATE(), INTERVAL 1 MONTH))
GROUP BY MONTHNAME(issue_date);
```

Result Output –

pmtd_loan_app
2279

2) Funded Amount (Total Loan Amount approved)

a) Total Funded Amount

```
select sum(loan_amount) as total_funded_Amount
from financial_loan;
```

Result Output –

total_funded_Amount
435757075

b) MTD Total Funded Amount

```
SELECT
SUM(loan_amount) AS mtd_total_funded_amount
from  financial_loan
where year(issue_date) = MONTH(curdate());
```

Result Output –

mtd_total_funded_amount
28875700

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c) PMTD Total Funded Amount

```
select
sum(loan_amount) as pmtd_total_funded_amount
from  financial_loan
where month(issue_date) = month(date_sub(curdate(), interval 1 month ))
group by monthname(issue_date)
```

Result Output –

pmtd_total_funded_amount
24647825

3) Amount Received(Loan Amount paid)

a) Total Amount Received

```
Select sum(total_Payment) as total_amount_received
 from  financial_loan;
```

Result Output –

total_amount_received
473070933

b) MTD Total Amount Received

```
select
sum(total_Payment) as mtd_total_amount_received
from financial_loan
where year(issue_date) = month(curdate());
```

Result Output –

mtd_total_amount_received
32264400

c) PMTD Total Amount Received

```
select
sum(total_Payment) as mtd_total_amount_received
from financial_loan
where year(issue_date) = month(date_sub(curdate(), interval 1 month ))
```

Result Output –

pmtd_total_amount_received
27717745

4) Interest Rate

a) Average Interest Rate

```
select avg(int_rate) * 100  as avg_int_rate
from  financial_loan;
```

Result Output –

avg_int_rate
12.04883142

b) MTD Average Interest

```
select
round(avg(int_rate) * 100,2)  as
mtd_avg_interest_rate from financial_loan
where month(issue_date) = MONTH(curdate());
```

Result Output –

mtd_avg_interest_rate
11.86

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c) PMTD Average Interest.

```
select
round(avg(int_rate) * 100,2)  as
mtd_avg_interest_rate from financial_loan
where month(issue_date) = month(date_sub(curdate(), interval 1 month));
```

Result Output –

pmtd_avg_interest_rate
11.72

5) DTI (Debt to Income ratio)

a) Avg DTI

```
select
round(avg(dti) *100,2) as
avg_dti From financial_loan;
```

Result Output –

avg_dti
13.32

b) MTD Avg DTI

```
select avg(dti) * 100 as mtd_avg_dti
from  financial_loan  where
month(issue_date) =
month(curdate());
```

Result Output –

mtd_avg_dti
13.66553779

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c) PMTD Avg DTI

```
SELECT
   ROUND(AVG(dti) * 100,2) AS pmtd_avg_dti
FROM financial_loan
WHERE MONTH(issue_date) = MONTH(DATE_SUB(CURDATE(), INTERVAL 1
MONTH));
```

Result Output –

pmtd_avg_dti
13.41

2. GOOD LOAN ISSUED

1. Good Loan Percentage

```
select (count(case
when loan_status = 'Fully Paid'  or loan_status = 'Current'
then id end) * 100.0) / count(id) as Good_Loan_Percentage
from financial_loan;
```

Result Output –

Good_Loan_Percentage
86.17534

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2. Good Loan Applications

```
select
count(id) as Good_Loan_Application
from  financial_loan
where  loan_status = 'Fully Paid' or loan_status = 'current';
```

Result Output –

Good_Loan_Application
33243

3. Good Loan Funded Amount

```
select  sum(loan_amount) as Good_Loan_Funded_Amount
FROM  financial_loan
where loan_status = 'Fully Paid' or loan_status = 'Current';
```

Result Output –

Good_Loan_Funded_Amount
370224850

4. Good Loan Amount Received

```
select sum(total_payment) as Good_Loan_Amount_Received
from financial_loan
where loan_status = 'Fully Paid'   or loan_status = 'Current';
```

Result Output –

Good_Loan_Amount_Received
435786170

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3. BAD LOAN ISSUED

1) . Bad Loan Percentage

```
SELECT
(count (case
when loan_status = 'Charged Off' THEN id
END) * 100.0) / COUNT(id) as Bad_Loan_Percentage
From financial_loan;
```

Result Output –

Bad_Loan_Percentage
13.82466

2). Bad Loan Applications

```
select
count(id) as Bad_Loan_Application
from
financial_loan
where loan_status = ' Charged Off  ';
```

Result Output –

Bad_Loan_Application
5333

3). Bad Loan Funded Amount

```
select
sum(loan_amount) as Bad_loan_Funded_Amount
from financial_loan
where  loan_status = 'Charged Off';
```

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Result Output –

Bad_loan_Funded_Amount
65532225

4). Bad Loan Amount Received

```
select
sum(total_payment) as Bad_Loan_Amount_Received
from
financial_loan
where   loan_status = 'Charged Off';
```

Result Output –

Bad_Loan_Amount_Received
37284763

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4. LOAN STATUS

1. Complete Loan Status Summary

```
select
loan_status,
count(id) as Total_Loan_Applications,
sum(total_payment) as Total_Amount_Received,
sum(loan_amount) as Total_Funded_Amount,
avg(int_rate * 100) as Interest_Rate,
avg(dti * 100) as DTI
from financial_loan
group by loan_status;
```

Result Output -

loan_status	Total_Loan_Applications	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
Fully Paid	32145	411586256	351358350	11.64	13.16
Charged Off	5333	37284763	65532225	13.88	14
Current	1098	24199914	18866500	15.1	14.72

2. MTD Loan Status Summary

```
select
loan_Status,
sum(loan_amount) AS MTD_Loan_Amount_Received,
sum(total_payment) as MTD_Loan_funded_Amound
from  financial_loan
where month(issue_date) = 12
group by loan_status;
```

Result Output –

loan_Status	MTD_Loan_Amount_Received	MTD_Loan_funded_Amound
Fully Paid	24791200	29870837
Charged Off	4075500	2381384
Current	9000	12179

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B. BANK LOAN REPORT | OVERVIEW

Showcase total number of applications, total loan amount and total amount received for the following parameters.

a. MONTH

```
select
month(issue_date) as Month_Number,
monthname(issue_date) as Month_Name,
count(id) as Total_Loan_Application,
sum(loan_amount) as Total_Loan_Funded_Amount,
sum(total_payment) as Total_Loan_Amount_Received
from financial_loan
group by  month(issue_date), monthname(issue_date)
order by  Month_Number;
```

Result Output –

Month_Number	Month_Name	Total_Loan_Application	Total_Loan_Funded_Amount	Total_Loan_Amount_Received
1	January	2332	25031650	27578836
2	February	2279	24647825	27717745
3	March	2627	28875700	32264400
4	April	2755	29800800	32495533
5	May	2911	31738350	33750523
6	June	3184	34161475	36164533
7	July	3366	35813900	38827220
8	August	3441	38149600	42682218
9	September	3536	40907725	43983948
10	October	3796	44893800	49399567
11	November	4035	47754825	50132030
12	December	4314	53981425	58074380

b. STATE

```
select address_state as state,
count(id) as Total_Loan_Application,
sum(loan_amount) as Total_Loan_Funded_Amount,
sum(total_payment) as Total_Loan_Amount_Received
from financial_loan
group by address_state
order by address_state;
```

Result Output – table is below

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state	Total_Loan_Application	Total_Loan_Funded_Amount	Total_Loan_Amount_Received
AK	78	1031800	1108570
AL	432	4949225	5492272
AR	236	2529700	2777875
AZ	833	9206000	10041986
CA	6894	78484125	83901234
CO	770	8976000	9845810
CT	730	8435575	9357612
DC	214	2652350	2921854
DE	110	1138100	1269136
FL	2773	30046125	31601905
GA	1355	15480325	16728040
HI	170	1850525	2080184
IA	5	56450	64482
ID	6	59750	65329
IL	1486	17124225	18875941
IN	9	86225	85521
KS	260	2872325	3247394
KY	320	3504100	3792530
LA	426	4498900	5001160
MA	1310	15051000	16676279
MD	1027	11911400	12985170
ME	3	9200	10808
MI	685	7829900	8543660
MN	592	6302600	6750746
MO	660	7151175	7692732
MS	19	139125	149342
MT	79	829525	892047
NC	759	8787575	9534813
NE	5	31700	24542
NH	161	1917900	2101386
NJ	1822	21657475	23425159
NM	183	1916775	2084485
NV	482	5307375	5451443
NY	3701	42077050	46108181
OH	1188	12991375	14330148
OK	293	3365725	3712649
OR	436	4720150	4966903
PA	1482	15826525	17462908
RI	196	1883025	2001774
SC	464	5080475	5462458
SD	63	606150	656514
TN	17	162175	141522
TX	2664	31236650	34392715
UT	252	2849225	2952412
VA	1375	15982650	17711443
VT	54	504100	534973
WA	805	8855525	9531739
WI	446	5070450	5485161
WV	167	1830525	1991936
WY	79	890750	1046050

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-- c. TERM

```
select
term as Term,
count(id) as Total_Loan_Application,
sum(loan_amount) as Total_Loan_Funded_Amount,
sum(total_payment) as Total_Loan_Amount_Received
from financial_loan
group by term
order by term;
```

Result Output –

Term	Total_Loan_Application	Total_Loan_Funded_Amount	Total_Loan_Amount_Received
36 months	28237	273041225	294709458
60 months	10339	162715850	178361475

-- d. EMPLOYEE LENGTH

```
select
emp_length as Employee_Length,
count(id) as Total_Loan_Application,
sum(loan_amount) as Total_Loan_Funded_Amount,
sum(total_payment) as Total_Loan_Amount_Received
from financial_loan
group by emp_length
order by emp_length;
```

Result Output –

Employee_Length	Total_Loan_Application	Total_Loan_Funded_Amount	Total_Loan_Amount_Received
< 1 year	4575	44210625	47545011
1 year	3229	32883125	35498348
10+ years	8870	116115950	125871616
2 years	4382	44967975	49206961
3 years	4088	43937850	47551832
4 years	3428	37600375	40964850
5 years	3273	36973625	40397571
6 years	2228	25612650	27908658
7 years	1772	20811725	22584136
8 years	1476	17558950	19025777
9 years	1255	15084225	16516173

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-- e. PURPOSE

```
select
purpose as Purpose,
count(id) as Total_Loan_Application,
sum(loan_amount) as Total_Loan_Funded_Amount,
sum(total_payment) as Total_Loan_Amount_Received
from financial_loan
group by purpose
order by purpose ;
```

Result Output –

Purpose	Total_Loan_Application	Total_Loan_Funded_Amount	Total_Loan_Amount_Received
car	1497	10223575	11324914
credit card	4998	58885175	65214084
Debt consolidation	18214	232459675	253801871
educational	315	2161650	2248380
home improvement	2876	33350775	36380930
house	366	4824925	5185538
major purchase	2110	17251600	18676927
medical	667	5533225	5851372
moving	559	3748125	3999899
other	3824	31155750	33289676
renewable_energy	94	845750	898931
small business	1776	24123100	23814817
vacation	352	1967950	2116738
wedding	928	9225800	10266856

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-- f. HOME OWNERSHIP

```
select
home_ownership as Home_Ownership,
count(id) as Total_Loan_Application,
sum(loan_amount) as Total_Loan_Funded_Amount,
sum(total_payment) as Total_Loan_Amount_Received
from financial_loan
group by home_ownership
order by home_ownership asc;
```

Result Output –

Home_Ownership	Total_Loan_Application	Total_Loan_Funded_Amount	Total_Loan_Amount_Received
MORTGAGE	17198	219329150	238474438
NONE	3	16800	19053
OTHER	98	1044975	1025257
OWN	2838	29597675	31729129
RENT	18439	185768475	201823056

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C. Miscellaneous | OVERVIEW

-- 1. MoM Loan Application growth rate

```
SELECT
ROUND((
  (
    SELECT COUNT(id)
    FROM financial_loan
    WHERE MONTH(issue_date) = MONTH(CURDATE())
  )
  -
  (
    SELECT COUNT(id)
    FROM financial_loan
    WHERE MONTH(issue_date) = MONTH(DATE_SUB(CURDATE(), INTERVAL 1
MONTH))
  )
) / NULLIF(
  (
    SELECT COUNT(id)
    FROM financial_loan
    WHERE MONTH(issue_date) = MONTH(DATE_SUB(CURDATE(), INTERVAL 1
MONTH))
  ), 0
) * 100,2) AS `MoM Total Applications Change (%)`;
```

Result Output –

MOM Total Applications Change (%)
15.27

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```
-- 2. Mom Loan Amount Disbursed growth rate
      SELECT
ROUND((
  (SELECT SUM(loan_amount)
   FROM financial_loan
   WHERE MONTH(issue_date) =
      MONTH(CURDATE()))
  -
  (SELECT SUM(loan_amount)
   FROM financial_loan
   WHERE MONTH(issue_date) =
      MONTH(DATE_SUB(CURDATE(),
      INTERVAL 1 MONTH)))
) / NULLIF(
  (SELECT SUM(loan_amount)
   FROM financial_loan
   WHERE MONTH(issue_date) =
      MONTH(DATE_SUB(CURDATE(),
      INTERVAL 1 MONTH))), 0
) * 100 ,2)AS `MoM Loan Amount Disbursed
      Growth Rate (%)`;
```

Result Output –

MoM Loan Amount Disbursed Growth Rate (%)
17.15

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-- 3. Interest rate for various subgrade and grade loan type

```
SELECT     a.grade,ROUND(b.avg_grade_rate * 100, 2) AS
Grade_interest, a.sub_grade,
       ROUND(AVG(a.int_rate) * 100, 2) AS Sub_Grade_interest
FROM financial_loan a
JOIN (
       SELECT grade, AVG(int_rate) AS avg_grade_rate
       FROM financial_loan
       GROUP BY grade
    ) b ON a.grade = b.grade
GROUP BY a.grade, a.sub_grade;
ORDER BY a.grade, a.sub_grade;
```

Result output :-

grade	Grade_interest	sub_grade	Sub_Grade_interest
A	7.35	A1	5.81
A	7.35	A2	6.39
A	7.35	A3	7.17
A	7.35	A4	7.66
A	7.35	A5	8.27
B	11.03	B1	10.01
B	11.03	B2	10.49
B	11.03	B3	11
B	11.03	B4	11.41
B	11.03	B5	11.79
C	13.55	C1	12.85
C	13.55	C2	13.35
C	13.55	C3	13.65
C	13.55	C4	14.04
C	13.55	C5	14.49
D	15.71	D1	14.84
D	15.71	D2	15.35
D	15.71	D3	15.76
D	15.71	D4	16.15
D	15.71	D5	16.63
E	17.71	E1	17.01
E	17.71	E2	17.43
E	17.71	E3	17.77
E	17.71	E4	18.26
E	17.71	E5	18.71
F	19.74	F1	19.2
F	19.74	F2	19.39
F	19.74	F3	20.03
F	19.74	F4	20.34
F	19.74	F5	20.73
G	21.4	G1	21.09
G	21.4	G2	21.07
G	21.4	G3	21.62
G	21.4	G4	21.97
G	21.4	G5	21.9

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Detailed Requirements:

"In order to monitor and assess our bank's lending activities and performance, we need to create a comprehensive Bank Loan Report. This report aims to provide insights into key loan-related metrics and their changes over time. The report will help us make data-driven decisions, track our loan portfolio's health, and identify trends that can inform our lending strategies.

Key Performance Indicators (KPIs) Requirements:

1. **Total Loan Applications:** We need to calculate the total number of loan applications received during a specified period. Additionally, it is essential to monitor the Month-to-Date (MTD) Loan Applications and track changes Month-over-Month (MoM).

2. **Total Funded Amount:** Understanding the total amount of funds disbursed as loans is crucial. We also want to keep an eye on the MTD Total Funded Amount and analyse the Month-over-Month (MoM) changes in this metric.

3. **Total Amount Received:** Tracking the total amount received from borrowers is essential for assessing the bank's cash flow and loan repayment. We should analyse the Month-to-Date (MTD) Total Amount Received and observe the Month-over-Month (MoM) changes.

4. **Average Interest Rate:** Calculating the average interest rate across all loans, MTD, and monitoring the Month-over-Month (MoM) variations in interest rates will provide insights into our lending portfolio's overall cost.

5. **Average Debt-to-Income Ratio (DTI):** Evaluating the average DTI for our borrowers helps us gauge their financial health. We need to compute the average DTI for all loans, MTD, and track Month-over-Month (MoM) fluctuations.

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Good Loan v Bad Loan KPI's

In order to evaluate the performance of our lending activities and assess the quality of our loan portfolio, we need to create a comprehensive report that distinguishes between 'Good Loans' and 'Bad Loans' based on specific loan status criteria

Good Loan KPIs:

1. **Good Loan Application Percentage:** We need to calculate the percentage of loan applications classified as 'Good Loans.' This category includes loans with a loan status of 'Fully Paid' and 'Current.'
2. **Good Loan Applications:** Identifying the total number of loan applications falling under the 'Good Loan' category, which consists of loans with a loan status of 'Fully Paid' and 'Current.'
3. **Good Loan Funded Amount:** Determining the total amount of funds disbursed as 'Good Loans.' This includes the principal amounts of loans with a loan status of 'Fully Paid' and 'Current.'
4. **Good Loan Total Received Amount:** Tracking the total amount received from borrowers for 'Good Loans,' which encompasses all payments made on loans with a loan status of 'Fully Paid' and 'Current.'

Bad Loan KPIs:

1. **Bad Loan Application Percentage:** Calculating the percentage of loan applications categorized as 'Bad Loans.' This category specifically includes loans with a loan status of 'Charged Off.'
2. **Bad Loan Applications:** Identifying the total number of loan applications categorized as 'Bad Loans,' which consists of loans with a loan status of 'Charged Off.'
3. **Bad Loan Funded Amount:** Determining the total amount of funds disbursed as 'Bad Loans.' This comprises the principal amounts of loans with a loan status of 'Charged Off.'

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4. **Bad Loan Total Received Amount:** Tracking the total amount received from borrowers for 'Bad Loans,' which includes all payments made on loans with a loan status of 'Charged Off.'

Loan Status Overview:

In order to gain a comprehensive overview of our lending operations and monitor the performance of loans, we aim to create a grid view report categorized by 'Loan Status.' This report will serve as a valuable tool for analysing and understanding the key indicators associated with different loan statuses. By providing insights into metrics such as 'Total Loan Applications,' 'Total Funded Amount,' 'Total Amount Received,' 'Month-to-Date (MTD) Funded Amount,' 'MTD Amount Received,' 'Average Interest Rate,' and 'Average Debt-to-Income Ratio (DTI),' this grid view will empower us to make data-driven decisions and assess the health of our loan portfolio.

1. Monthly Trends by Issue Date:

Metrics: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Objective: This will showcase how 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received' vary over time, allowing us to identify seasonality and long-term trends in lending activities.

2. Regional Analysis by State:

Metrics: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Objective: This will represent lending metrics categorized by state, enabling us to identify regions with significant lending activity and assess regional disparities.

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3. Loan Term Analysis:

Metrics: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Segments: Loan Terms (e.g., 36 months, 60 months)

Objective: This will depict loan statistics based on different loan terms, allowing us to understand the distribution of loans across various term lengths.

4. Employee Length Analysis:

Metrics: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Objective: This will illustrate how lending metrics are distributed among borrowers with different employment lengths, helping us assess the impact of employment history on loan applications.

5. Loan Purpose Breakdown:

Metrics: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Objective: This will provide a breakdown of loan metrics based on the stated purposes of loans, aiding in the understanding of the primary reasons borrowers seek financing.

6. Home Ownership Analysis:

Metrics: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Hierarchy: Home Ownership Categories (e.g., own, rent, mortgage)

********** **The End** **********

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